DATABASE MANAGEMENT SYSTEM LAB

Name: Aditi M

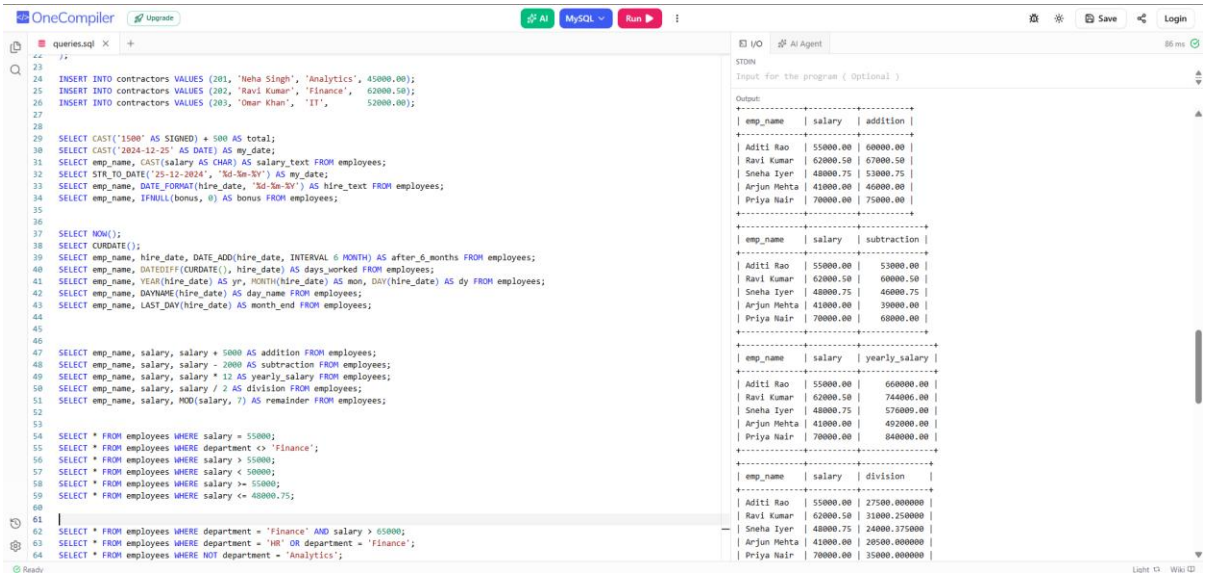
USN: 1RUA25BCA0006

Subject: Database Management System

LAB 5

Question: Implement SQL Operators: Demonstrate the use of different types of operators in SQL with examples, including Arithmetic Operators, Logical Operators, Comparison Operators.

Output:



```
INSERT INTO contractors VALUES (201, 'Neha Singh', 'Analytics', 45000.00);
INSERT INTO contractors VALUES (202, 'Ravi Kumar', 'Finance',   62000.50);
INSERT INTO contractors VALUES (203, 'Omar Khan',  'IT',        52000.00);


SELECT CAST('1500' AS SIGNED) + 500 AS total;
SELECT CAST('2024-12-25' AS DATE) AS my_date;
SELECT emp_name, CAST(salary AS CHAR) AS salary_text FROM employees;
SELECT STR_TO_DATE('25-12-2024', '%d-%m-%Y') AS my_date;
SELECT emp_name, DATE_FORMAT(hire_date, '%d-%m-%Y') AS hire_text FROM employees;
SELECT emp_name, IFNULL(bonus, 0) AS bonus FROM employees;


SELECT NOW();
SELECT CURDATE();
SELECT emp_name, hire_date, DATE_ADD(hire_date, INTERVAL 6 MONTH) AS after_6_months FROM employees;
SELECT emp_name, DATEDIFF(CURDATE(), hire_date) AS days_worked FROM employees;
SELECT emp_name, YEAR(hire_date) AS yr, MONTH(hire_date) AS mon, DAY(hire_date) AS dy FROM employees;
SELECT emp_name, DAYNAME(hire_date) AS day_name FROM employees;
SELECT emp_name, LAST_DAY(hire_date) AS month_end FROM employees;



SELECT emp_name, salary, salary + 5000 AS addition FROM employees;
SELECT emp_name, salary, salary - 2000 AS subtraction FROM employees;
SELECT emp_name, salary, salary * 12 AS yearly_salary FROM employees;
SELECT emp_name, salary, salary / 2 AS division FROM employees;
SELECT emp_name, salary, MOD(salary, 7) AS remainder FROM employees;


SELECT * FROM employees WHERE salary = 55000;
SELECT * FROM employees WHERE department <> 'Finance';
SELECT * FROM employees WHERE salary > 55000;
SELECT * FROM employees WHERE salary < 50000;
SELECT * FROM employees WHERE salary >= 55000;
SELECT * FROM employees WHERE salary <= 48000.75;

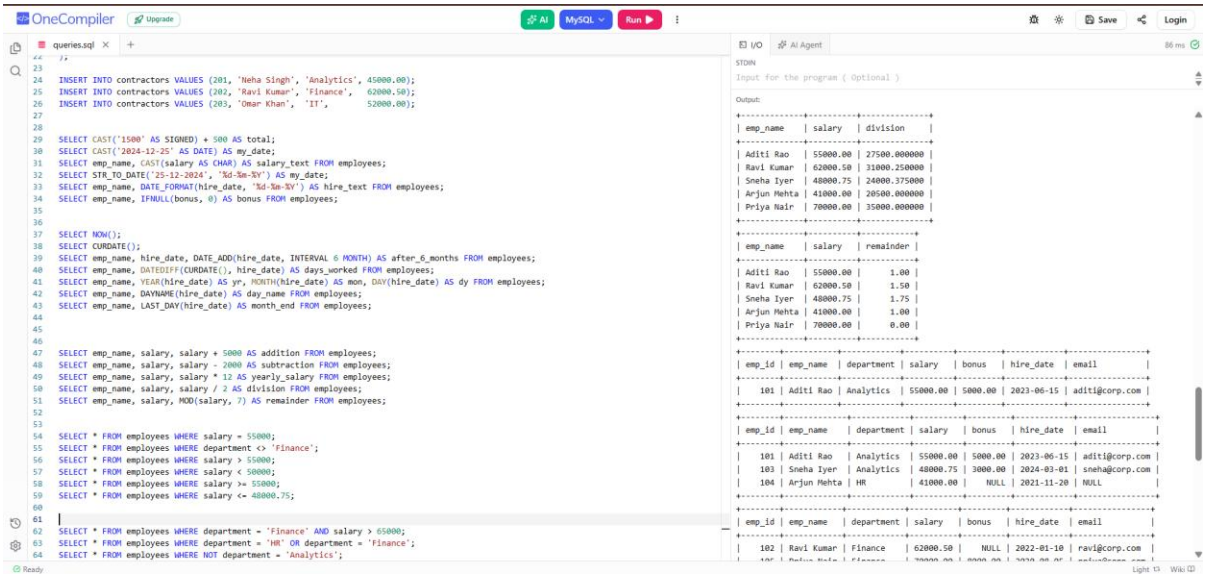

SELECT * FROM employees WHERE department = 'Finance' AND salary > 65000;
SELECT * FROM employees WHERE department = 'HR' OR department = 'Finance';
SELECT * FROM employees WHERE NOT department = 'Analytics';
```

```
+------------+---------+----------+
| emp_name   | salary  | addition |
+------------+---------+----------+
| Aditi Rao  | 55000.00 | 60000.00 |
| Ravi Kumar | 62000.50 | 67000.50 |
| Sneha Iyer | 48000.75 | 53000.75 |
| Arjun Mehta | 41000.00 | 46000.00 |
| Priya Nair | 70000.00 | 75000.00 |
+------------+---------+----------+

+------------+---------+-------------+
| emp_name   | salary  | subtraction |
+------------+---------+-------------+
| Aditi Rao  | 55000.00 |    53000.00 |
| Ravi Kumar | 62000.50 |    60000.50 |
| Sneha Iyer | 48000.75 |    46000.75 |
| Arjun Mehta | 41000.00 |    39000.00 |
| Priya Nair | 70000.00 |    68000.00 |
+------------+---------+-------------+

+------------+---------+--------------+
| emp_name   | salary  | yearly_salary |
+------------+---------+--------------+
| Aditi Rao  | 55000.00 |     660000.00 |
| Ravi Kumar | 62000.50 |     744006.00 |
| Sneha Iyer | 48000.75 |     576009.00 |
| Arjun Mehta | 41000.00 |     492000.00 |
| Priya Nair | 70000.00 |     840000.00 |
+------------+---------+--------------+

+------------+---------+-------------+
| emp_name   | salary  | division    |
+------------+---------+-------------+
| Aditi Rao  | 55000.00 | 27500.000000 |
| Ravi Kumar | 62000.50 | 31000.250000 |
| Sneha Iyer | 48000.75 | 24000.375000 |
| Arjun Mehta | 41000.00 | 20500.000000 |
| Priya Nair | 70000.00 | 35000.000000 |
```




```
);

INSERT INTO contractors VALUES (201, 'Neha Singh', 'Analytics', 45000.00);
INSERT INTO contractors VALUES (202, 'Ravi Kumar', 'Finance',   62000.50);
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SELECT CAST('1500' AS SIGNED) + 500 AS total;
SELECT CAST('2024-12-25' AS DATE) AS my_date;
SELECT emp_name, CAST(salary AS CHAR) AS salary_text FROM employees;
SELECT STR_TO_DATE('25-12-2024', '%d-%m-%Y') AS my_date;
SELECT emp_name, DATE_FORMAT(hire_date, '%d-%m-%Y') AS hire_text FROM employees;
SELECT emp_name, IFNULL(bonus, 0) AS bonus FROM employees;


SELECT NOW();
SELECT CURDATE();
SELECT emp_name, hire_date, DATE_ADD(hire_date, INTERVAL 6 MONTH) AS after_6_months FROM employees;
SELECT emp_name, DATEDIFF(CURDATE(), hire_date) AS days_worked FROM employees;
SELECT emp_name, YEAR(hire_date) AS yr, MONTH(hire_date) AS mon, DAY(hire_date) AS dy FROM employees;
SELECT emp_name, DAYNAME(hire_date) AS day_name FROM employees;
SELECT emp_name, LAST_DAY(hire_date) AS month_end FROM employees;



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SELECT emp_name, salary, salary - 2000 AS subtraction FROM employees;
SELECT emp_name, salary, salary * 12 AS yearly_salary FROM employees;
SELECT emp_name, salary, salary / 2 AS division FROM employees;
SELECT emp_name, salary, MOD(salary, 7) AS remainder FROM employees;


SELECT * FROM employees WHERE salary = 55000;
SELECT * FROM employees WHERE department <> 'Finance';
SELECT * FROM employees WHERE salary > 55000;
SELECT * FROM employees WHERE salary < 50000;
SELECT * FROM employees WHERE salary >= 55000;
SELECT * FROM employees WHERE salary <= 48000.75;


SELECT * FROM employees WHERE department = 'Finance' AND salary > 65000;
SELECT * FROM employees WHERE department = 'HR' OR department = 'Finance';
SELECT * FROM employees WHERE NOT department = 'Analytics';
```

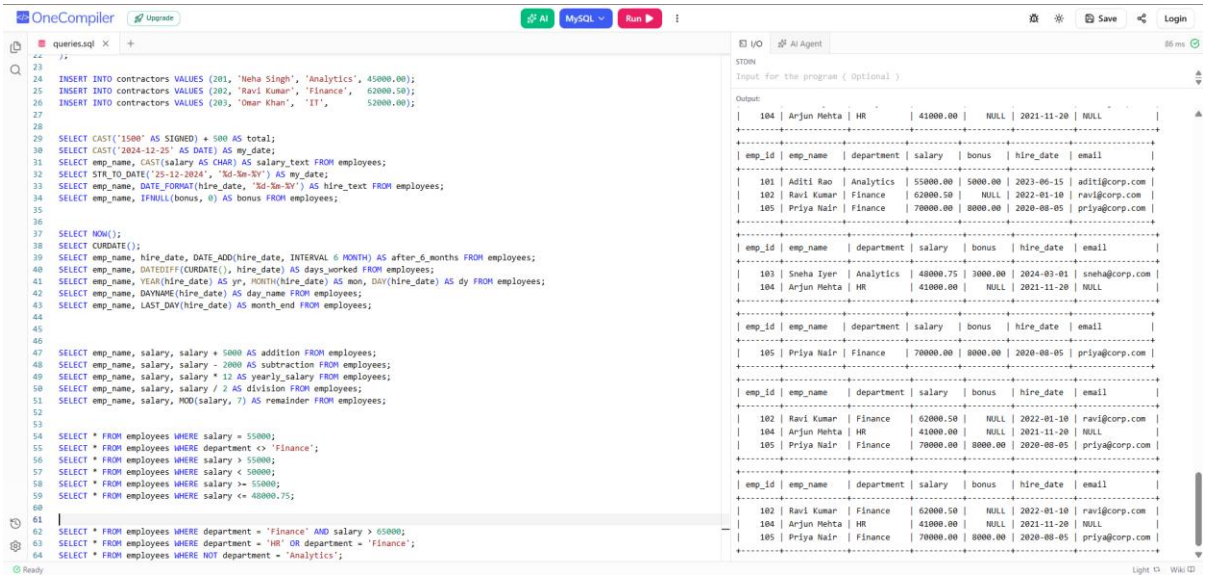
```
+-------------+----------+-------------+
| emp_name    | salary   | division    |
+-------------+----------+-------------+
| Aditi Rao   | 55000.00 | 27500.000000 |
| Ravi Kumar  | 62000.50 | 31000.250000 |
| Sneha Iyer  | 48000.75 | 24000.375000 |
| Arjun Mehta | 41000.00 | 20500.000000 |
| Priya Nair  | 70000.00 | 35000.000000 |
+-------------+----------+-------------+

+-------------+----------+-----------+
| emp_name    | salary   | remainder |
+-------------+----------+-----------+
| Aditi Rao   | 55000.00 |      1.00 |
| Ravi Kumar  | 62000.50 |      1.50 |
| Sneha Iyer  | 48000.75 |      1.75 |
| Arjun Mehta | 41000.00 |      1.00 |
| Priya Nair  | 70000.00 |      0.00 |
+-------------+----------+-----------+

+--------+-------------+------------+----------+---------+------------+-----------------+
| emp_id | emp_name    | department | salary   | bonus   | hire_date  | email           |
+--------+-------------+------------+----------+---------+------------+-----------------+
|    101 | Aditi Rao   | Analytics  | 55000.00 | 5000.00 | 2023-06-15 | aditi@corp.com  |
+--------+-------------+------------+----------+---------+------------+-----------------+

+--------+-------------+------------+----------+---------+------------+-----------------+
| emp_id | emp_name    | department | salary   | bonus   | hire_date  | email           |
+--------+-------------+------------+----------+---------+------------+-----------------+
|    101 | Aditi Rao   | Analytics  | 55000.00 | 5000.00 | 2023-06-15 | aditi@corp.com  |
|    103 | Sneha Iyer  | Analytics  | 48000.75 | 3000.00 | 2024-03-01 | sneha@corp.com  |
|    104 | Arjun Mehta | HR         | 41000.00 |    NULL | 2021-11-20 | NULL            |
+--------+-------------+------------+----------+---------+------------+-----------------+

+--------+-------------+------------+----------+---------+------------+-----------------+
| emp_id | emp_name    | department | salary   | bonus   | hire_date  | email           |
+--------+-------------+------------+----------+---------+------------+-----------------+
|    102 | Ravi Kumar  | Finance    | 62000.50 |    NULL | 2022-01-10 | ravi@corp.com   |
```




```
);

INSERT INTO contractors VALUES (201, 'Neha Singh', 'Analytics', 45000.00);
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INSERT INTO contractors VALUES (203, 'Omar Khan',  'IT',        52000.00);


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SELECT emp_name, IFNULL(bonus, 0) AS bonus FROM employees;


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SELECT * FROM employees WHERE salary = 55000;
SELECT * FROM employees WHERE department <> 'Finance';
SELECT * FROM employees WHERE salary > 55000;
SELECT * FROM employees WHERE salary < 50000;
SELECT * FROM employees WHERE salary >= 55000;
SELECT * FROM employees WHERE salary <= 48000.75;


SELECT * FROM employees WHERE department = 'Finance' AND salary > 65000;
SELECT * FROM employees WHERE department = 'HR' OR department = 'Finance';
SELECT * FROM employees WHERE NOT department = 'Analytics';
```

```
|    104 | Arjun Mehta | HR         | 41000.00 |    NULL | 2021-11-20 | NULL            |
+--------+-------------+------------+----------+---------+------------+-----------------+

+--------+-------------+------------+----------+---------+------------+-----------------+
| emp_id | emp_name    | department | salary   | bonus   | hire_date  | email           |
+--------+-------------+------------+----------+---------+------------+-----------------+
|    101 | Aditi Rao   | Analytics  | 55000.00 | 5000.00 | 2023-06-15 | aditi@corp.com  |
|    102 | Ravi Kumar  | Finance    | 62000.50 |    NULL | 2022-01-10 | ravi@corp.com   |
|    105 | Priya Nair  | Finance    | 70000.00 | 8000.00 | 2020-08-05 | priya@corp.com  |
+--------+-------------+------------+----------+---------+------------+-----------------+

+--------+-------------+------------+----------+---------+------------+-----------------+
| emp_id | emp_name    | department | salary   | bonus   | hire_date  | email           |
+--------+-------------+------------+----------+---------+------------+-----------------+
|    103 | Sneha Iyer  | Analytics  | 48000.75 | 3000.00 | 2024-03-01 | sneha@corp.com  |
|    104 | Arjun Mehta | HR         | 41000.00 |    NULL | 2021-11-20 | NULL            |
+--------+-------------+------------+----------+---------+------------+-----------------+

+--------+-------------+------------+----------+---------+------------+-----------------+
| emp_id | emp_name    | department | salary   | bonus   | hire_date  | email           |
+--------+-------------+------------+----------+---------+------------+-----------------+
|    105 | Priya Nair  | Finance    | 70000.00 | 8000.00 | 2020-08-05 | priya@corp.com  |
+--------+-------------+------------+----------+---------+------------+-----------------+

+--------+-------------+------------+----------+---------+------------+-----------------+
| emp_id | emp_name    | department | salary   | bonus   | hire_date  | email           |
+--------+-------------+------------+----------+---------+------------+-----------------+
|    102 | Ravi Kumar  | Finance    | 62000.50 |    NULL | 2022-01-10 | ravi@corp.com   |
|    104 | Arjun Mehta | HR         | 41000.00 |    NULL | 2021-11-20 | NULL            |
|    105 | Priya Nair  | Finance    | 70000.00 | 8000.00 | 2020-08-05 | priya@corp.com  |
+--------+-------------+------------+----------+---------+------------+-----------------+

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|    102 | Ravi Kumar  | Finance    | 62000.50 |    NULL | 2022-01-10 | ravi@corp.com   |
|    104 | Arjun Mehta | HR         | 41000.00 |    NULL | 2021-11-20 | NULL            |
|    105 | Priya Nair  | Finance    | 70000.00 | 8000.00 | 2020-08-05 | priya@corp.com  |
+--------+-------------+------------+----------+---------+------------+-----------------+
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